

MUGUNTHAN KESAVAN

AI / ML ENGINEER — LLM SYSTEMS & GENERATIVE AI

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SUMMARY

AI/ML engineer building production LLM systems — multi-stage model pipelines, retrieval-augmented generation, and deployment. Currently delivering an order-validation pipeline and a client-facing RAG assistant for a manufacturing client. Two peer-reviewed publications. M.S. candidate,

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, OpenCV, Deep Learning, NLP, Computer Vision

LLM Engineering: Harness Engineering, RAG, Agentic Workflows, Multi-Agent Systems, Prompt Engineering, Fine-Tuning, Embeddings, Vector Search, LLM Evaluation

Agents & Protocols: LangChain, LangGraph, Google ADK, MCP, A2A, MHS, CrewAI, Firecrawl

Digital Twins & Infrastructure: Digital Twin Architecture, Discrete-Event Simulation, SimPy, Process Modeling, What-If Analysis, Bottleneck Analysis, Docker, FastAPI, CI/CD, Git

Data & Tools: PostgreSQL, pgvector, Qdrant, ChromaDB, MongoDB, Spark, Pandas, NumPy, Vertex AI, Supabase, Streamlit, SHAP, LIME

Languages: Python, SQL, Java

EXPERIENCE

Technical Research Assistant — AI Systems & Automation Mar 2026 – Present

Texas Manufacturing Assistance Center (TMAC) x University of Houston — Houston, TX

- Architected DeltaZero, a two-stage order-validation pipeline: Llama 3.1 8B for error detection with a 70B judge model, confidence-tiered routing, and deterministic normalization as a pre-LLM filter — 97% detection accuracy against a manually labeled ground-truth set.
- Reduced order-entry errors from a client-observed baseline of roughly 20% to under 1% on evaluated samples; running precision tuning ahead of a 350-sample stratified evaluation.
- Built a client-facing RAG assistant over 3,500+ pages of technical and financial documentation, implementing recursive chunking, vector retrieval, and cross-encoder reranking to improve evidence selection and answer relevance.
- Containerized the stack with Docker and set up CI/CD for reproducible deployment and clean handover across environments and team members.

Research Assistant — Digital Twin & Optimization Jan 2026 – May 2026

Hewlett Packard Enterprise (HPE) x University of Houston — Houston, TX

- Led end-to-end development of an 8-station reverse-supply-chain digital twin using discrete-event simulation (SimPy), enabling bottleneck detection through dwell-time analysis and what-if scenario modeling at an estimated 60x simulation speed.
- Engineered the full-stack system (Python, FastAPI, React 19, Three.js, SQLite) with an LLM-powered anomaly-diagnosis assistant using Ollama (Phi-3 Mini) for natural-language Q&A over operational KPIs.
- Automated unit tracking with QR codes, an estimated 90% reduction in manual tracking effort; presented to HPE leadership identifying potential for 15%+ throughput improvement.

Research Intern — Machine Learning Jan 2025 – Jul 2025

Flame University — Pune, India

- Applied transfer learning with transformer encoders (ProtBERT, ESM2) and SVM classifiers across 30,000+ protein sequences using 10-fold cross-validation, reaching 98.3% accuracy and 98% F1.
- Authored a peer-reviewed Springer publication; automated data preprocessing in Python/Pandas, cutting manual effort 50%.

EDUCATION

M.S., Engineering Data Science & Artificial Intelligence — GPA 3.8/4.0 Aug 2025 – May 2027

University of Houston — Houston, TX

Coursework: Machine Learning, Data Science, Statistics, Neural Networks, Cloud Computing(AWS,GCP),Biomedical Data Mining, Predictive Analytics

B.Tech., Artificial Intelligence Engineering — GPA 3.7/4.0 Aug 2021 – May 2025

Amrita Vishwa Vidyapeetham — India

Coursework: Machine Learning, Deep Learning, Big Data Analytics, Natural Language Processing, Distributed Systems

SELECTED PROJECTS

- Segmented Neural Style Transfer for Real-Time Video — region-specific styling via segmentation, with a temporal loss for frame coherence and TensorBoard-tracked evaluation. github.com/mugunthank7/Enhanced-Segmented-Neural-Style-Transfer-for-Real-Time-Video-Stylization
- NYC TLC Data Engineering Pipeline — automated scraping, parquet consolidation, reproducible sampling, feature engineering, and KMeans clustering. github.com/mugunthank7/nyc-tlc-data-engineering-pipeline

PUBLICATIONS

"Humor Detection in Telugu Social Media Text Using Cost-Sensitive Learning." ICRTC 2024.

"Comparative Study of Zero-Shot and Supervised Learning for Tamil Tweet Sentiment Analysis." ICDSA 2025 (Springer).

LEADERSHIP & ACTIVITIES

Led Python/ML workshops for 70+ students | Mentored 10+ juniors on AI/DS projects | INFORMS member